Endogeneity and Moment-Based Estimation

From OLS Failure to IV, 2SLS, GMM, Diagnostics, and Monte Carlo Evidence

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Table of contents

List of Figures

List of Tables

1. Introduction, Motivation, and Roadmap

This handout examines one of the central problems in applied econometrics: the estimation of structural relationships when one or more regressors are endogenous. In the classical linear regression model, ordinary least squares (OLS) provides reliable estimates only when the regressors are orthogonal to the disturbance term. When this orthogonality condition fails, OLS is no longer centered on the true structural parameter, even asymptotically. The main objective of this handout is therefore to explain why endogeneity invalidates OLS, how instrumental variables restore identification, and how IV, 2SLS, and GMM estimators provide consistent and, under suitable conditions, efficient alternatives.

The discussion begins from the classical linear model because it provides the benchmark against which endogeneity is understood. Under standard assumptions, especially the condition $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$, OLS is consistent and asymptotically normal. Once this moment condition is violated, however, the probability limit of OLS differs from the true parameter vector. Endogeneity may arise from omitted variables, simultaneity, measurement error, sample selection, dynamic feedback, or other forms of misspecification. In all cases, the essential problem is the same: some component of the regressor contains variation that is systematically related to the unobserved disturbance. Hence, endogeneity is fundamentally an identification problem, not merely a problem of small samples or imprecise estimation.

The handout develops three related solutions. The first is the simple instrumental variables estimator, which replaces the invalid regressor-error orthogonality condition with a valid instrument-error orthogonality condition. A valid instrument must be relevant for the endogenous regressor and exogenous with respect to the structural error. The second is two-stage least squares, which extends the IV logic to models with multiple instruments by projecting the endogenous regressors onto the space spanned by the instruments and then estimating the structural equation using the projected variation. The third is the generalized method of moments, which nests IV and 2SLS as special cases and estimates parameters by minimizing a quadratic form in the sample moment conditions. Particular attention is given to consistency, asymptotic normality, efficiency, and the role of the GMM weighting matrix.

The roadmap of the handout is as follows. Section~2 introduces the IV estimator, derives its matrix form, and establishes its consistency under instrument relevance and exogeneity. Section~3 develops the 2SLS estimator, explains its first-stage and second-stage interpretation, and discusses its asymptotic properties and variance estimation under homoskedasticity, heteroskedasticity, and serial correlation. Section~3.3 presents key diagnostic tests, including tests for endogeneity, weak instruments, and overidentifying restrictions. Section~4 introduces GMM, derives the linear GMM estimator, explains the optimal weighting matrix, and presents consistency, asymptotic normality, and efficiency results. The final sections synthesize the main ideas, provide review exercises, and include a Monte Carlo appendix illustrating the finite-sample behaviour of OLS, simple IV, 2SLS, and GMM when one regressor is endogenous.

1.1 The Classical Linear Model and Its Assumptions

Let $\{(y_i, \mathbf{x}_i)\}_{i=1}^n$ be a sample of n observations, where $y_i \in \mathbb{R}$ is the dependent variable and $\mathbf{x}_i \in \mathbb{R}^k$ is a vector of regressors, with the convention that its first component equals unity. The *classical linear model* (CLM) postulates y 1

$$y_i \;=\; \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i, \qquad i = 1, \dots, n, \tag{1}$$

where $\boldsymbol{\beta}_0 \in \mathbb{R}^k$ is the vector of true parameters and ε_i is an unobservable disturbance. Written in matrix form, (1) becomes $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$, where $\mathbf{Y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n\times k}$, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ collect the observations in the obvious way.

The ordinary least-squares (OLS) estimator is

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} \;=\; (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \;=\; \boldsymbol{\beta}_0 + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}. \tag{2}$$

Its desirable properties depend on the following assumptions, which we state in a form suitable for asymptotic analysis.

Assumption 1.1 (Linearity)**.** The data-generating process satisfies (1) for some $\boldsymbol{\beta}_0 \in \mathbb{R}^k$.

Assumption 1.2 (Random Sampling)**.** $\{(y_i, \mathbf{x}_i)\}_{i=1}^n$ is an independent and identically distributed (i.i.d.) sequence drawn from a well-defined population distribution.

Assumption 1.3 (Full Rank)**.** $\mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top]$ is finite and positive definite, so that $\operatorname{rank}(\mathbf{X}'\mathbf{X}) = k$ almost surely for n sufficiently large.

Assumption 1.4 (Strict Exogeneity)**.** $\mathbb{E}[\varepsilon_i \mid \mathbf{x}_i] = 0$ almost surely. As an immediate consequence, $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$ and $\mathbb{E}[\varepsilon_i] = 0$.

Assumption 1.5 (Spherical Disturbances)**.** $\mathbb{E}[\varepsilon_i^2 \mid \mathbf{x}_i] = \sigma^2 \in (0, \infty)$ and $\mathbb{E}[\varepsilon_i\varepsilon_j \mid \mathbf{x}_i, \mathbf{x}_j] = 0$ for all $i \neq j$.

Remark 1.6. Assumption 1.4 is the pivot around which the entire analysis turns. As stated, it is a *mean independence* condition: $\mathbb{E}[\varepsilon_i \mid \mathbf{x}_i] = 0$. This condition implies the weaker orthogonality condition $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$, but the converse does not hold in general. For consistency of OLS, the orthogonality condition is the key requirement. Stronger conditional mean assumptions are mainly useful for deriving sharper finite-sample or conditional results.

Under Assumptions~1.1–1.5, the Gauss–Markov theorem establishes that OLS is the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}_0$. Consistency and asymptotic normality follow from a standard application of the law of large numbers and the central limit theorem; see Wooldridge [16], Hayashi [11], and Hansen [10].

Theorem 1.7 (Consistency and Asymptotic Normality of OLS)**.** *Under Assumptions 1.1–1.4 and the moment condition* $\mathbb{E}\left[\|\mathbf{x}_i\|^2 \varepsilon_i^2\right] < \infty$*,*

(i) $\hat{\boldsymbol{\beta}}_{\text{OLS}} \xrightarrow{p} \boldsymbol{\beta}_0$*; and*

(ii) $\sqrt{n}\,(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0},\, \boldsymbol{\Sigma}_{\text{OLS}})$*, where* $\boldsymbol{\Sigma}_{\text{OLS}} = \left(\mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top]\right)^{-1} \mathbb{E}[\varepsilon_i^2 \mathbf{x}_i\mathbf{x}_i^\top] \left(\mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top]\right)^{-1}$.

Under the additional Assumption 1.5, $\boldsymbol{\Sigma}_{\text{OLS}}$ *simplifies to* $\sigma^2\left(\mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top]\right)^{-1}$.

Proof. From (2),

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}_0 \;=\; \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1} \frac{1}{n}\mathbf{X}'\boldsymbol{\varepsilon}.$$

By the LLN (Assumption 1.2), $n^{-1}\mathbf{X}'\mathbf{X} \xrightarrow{p} \mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top]$, which is invertible by Assumption 1.3, and $n^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \xrightarrow{p} \mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$ by Assumption 1.4. Part (i) follows from Slutsky's theorem. For part (ii), write $n^{-1/2}\mathbf{X}'\boldsymbol{\varepsilon} = n^{-1/2}\sum_{i=1}^n \mathbf{x}_i\varepsilon_i$. The summands are i.i.d. with mean zero and covariance matrix $\mathbb{E}[\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i^\top]$; the CLT yields $n^{-1/2}\mathbf{X}'\boldsymbol{\varepsilon} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbb{E}[\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i^\top])$. The delta method (or Slutsky) completes the argument. $■$

1.2 Endogeneity: Definition, Sources, and Consequences

The conditional mean assumption in Assumption~1.4 is sufficient but stronger than necessary for OLS consistency. The key failure for consistency occurs when the weaker orthogonality condition $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$ is violated.

Definition 1.8 (Endogeneity)**.** A regressor x_{ji} (the j-th component of $\mathbf{x}_i$) is said to be *endogenous* if $\text{Cov}(x_{ji}, \varepsilon_i) \neq 0$, or, more generally, if $\mathbb{E}[\mathbf{x}_i\varepsilon_i] \neq \mathbf{0}$. The model (1) is then said to suffer from *endogeneity*; compare Wooldridge [16], Cameron and Trivedi [2], and Hansen [10].

Endogeneity is not a single pathology but a family of related problems sharing a common algebraic signature. We distinguish three principal sources.

1.2.1 Omitted Variable Bias

Suppose the true model is

$$y_i \;=\; \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \gamma_0 c_i + u_i, \qquad \mathbb{E}[u_i \mid \mathbf{x}_i, c_i] = 0,$$

where c_i is an unobserved confounder correlated with $\mathbf{x}_i$. If we estimate the misspecified model (1) with $\varepsilon_i = \gamma_0 c_i + u_i$, then $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \gamma_0\,\mathbb{E}[\mathbf{x}_i c_i] \neq \mathbf{0}$ whenever $\gamma_0 \neq 0$ and c_i is correlated with at least one regressor. This is the *omitted variable bias* (OVB) problem, arguably the most common form of endogeneity in applied economics; see Wooldridge [16] and Cameron and Trivedi [2].

1.2.2 Simultaneity Bias

In many economic settings, y_i and one or more components of $\mathbf{x}_i$ are determined jointly by a system of equations. Hayashi [11] and Davidson and MacKinnon [7] discuss the canonical supply-and-demand example:

$$\begin{aligned}
q_i^d &= \alpha_0 + \alpha_1 p_i + \alpha_2 \mathbf{w}_i' \boldsymbol{\delta} + u_i^d \quad \text{(demand)},\\
q_i^s &= \beta_0 + \beta_1 p_i + \beta_2 \mathbf{r}_i' \boldsymbol{\phi} + u_i^s \quad \text{(supply)},\\
q_i^d &= q_i^s \quad \text{(market clearing)}.
\end{aligned}$$

In equilibrium, price p_i is a function of both structural errors u_i^d and u_i^s. Regressing quantity on price therefore yields an estimator that is inconsistent for either structural parameter.

1.2.3 Measurement Error

Let x_i^* be the true value of a regressor, observed with error: $x_i = x_i^* + v_i$, where v_i is classical measurement error with $\mathbb{E}[v_i] = 0$, $\operatorname{Cov}(x_i^*, v_i) = 0$, and $\operatorname{Cov}(u_i, v_i) = 0$. Assume also that the true regressor is exogenous, so that $\operatorname{Cov}(x_i^*, u_i) = 0$. As Wooldridge [16] and Cameron and Trivedi [2] show, substituting into the true model $y_i = \beta_0 + \beta_1 x_i^* + u_i$ gives

$$y_i \;=\; \beta_0 + \beta_1 x_i + \underbrace{(u_i - \beta_1 v_i)}_{=\varepsilon_i}.$$

Since $\operatorname{Cov}(x_i, \varepsilon_i) = -\beta_1 \operatorname{Var}(v_i) \neq 0$, OLS is inconsistent. The resulting *attenuation bias* pushes the estimate of β_1 toward zero in absolute value.

The consequences of endogeneity are stark.

Theorem 1.9 (Inconsistency of OLS under Endogeneity)**.** *Suppose Assumption 1.4 fails, so that* $\boldsymbol{\delta} \equiv \mathbb{E}[\mathbf{x}_i \varepsilon_i] \neq \mathbf{0}$*. Then, under Assumptions 1.1–1.3,*

$$\operatorname{plim}_{n\to\infty} \hat{\boldsymbol{\beta}}_{\text{OLS}} \;=\; \boldsymbol{\beta}_0 + \left(\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]\right)^{-1} \boldsymbol{\delta} \;\neq\; \boldsymbol{\beta}_0.$$

Moreover, no amount of additional data can eliminate this asymptotic bias*.*

Proof. By the LLN applied to (2),

$$\operatorname{plim} \hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta}_0 + \left(\operatorname{plim} \frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \left(\operatorname{plim} \frac{\mathbf{X}'\boldsymbol{\varepsilon}}{n}\right) = \boldsymbol{\beta}_0 + \left(\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]\right)^{-1} \boldsymbol{\delta}.$$

Because $\boldsymbol{\delta} \neq \mathbf{0}$ by hypothesis and $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]$ is non-singular by Assumption 1.3, the second term is non-zero and persists as $n \to \infty$. $■$

Remark 1.10. Theorem 1.9 makes clear that endogeneity is fundamentally an identification problem, not a finite-sample one. Collecting more data merely produces a more precise estimate of the wrong quantity. The solution must address the source of the correlation $\boldsymbol{\delta}$, not the sample size.

Example 1.11 (Returns to Schooling)**.** Consider the classic Mincer earnings equation,

$$\log w_i \;=\; \beta_0 + \beta_1 S_i + \mathbf{x}_i^\top \boldsymbol{\gamma} + \varepsilon_i,$$

where w_i is the wage, S_i is years of schooling, and $\mathbf{x}_i$ contains observable controls (age, region, etc.). The parameter of interest, β_1, measures the average causal return to an additional year of education. The problem is that S_i is almost certainly correlated with unobserved *ability* a_i, which also affects wages and is absorbed into ε_i. If ability raises both schooling and wages, OLS will tend to overstate β_1. Card [3] addresses this by instrumenting S_i with proximity to a college, a variable that plausibly affects schooling attainment without directly influencing wages — an application of the IV strategy introduced in Section 2.

1.3 A Preview of Solutions

Three broad strategies have been developed to recover consistent estimates of $\boldsymbol{\beta}_0$ in the presence of endogeneity. These notes treat the first two in depth and briefly touch on their connections to the third.

Instrumental Variables (IV). The IV approach introduces a vector of *instruments* $\mathbf{z}_i \in \mathbb{R}^\ell$ that satisfy two conditions: *relevance*, expressed in the multivariate model by the rank condition $\text{rank}\{\mathbb{E}[\mathbf{z}_i\mathbf{x}_i^\top]\} = k$, and *exogeneity*, $\mathbb{E}[\mathbf{z}_i\varepsilon_i] = \mathbf{0}$. As Hayashi [11], Hansen [10], and Wooldridge [16] explain, these two conditions together restore identification. Section~2 develops the IV estimator, establishes its asymptotic properties, and discusses the critical practical issue of *instrument strength*.

Two-Stage Least Squares (2SLS). When $\ell > k$ — i.e. when the number of instruments exceeds the number of structural parameters — the model is *overidentified* and the sample moment equations cannot generally all be set to zero. Equivalently, when included exogenous regressors instrument themselves, overidentification means that the number of excluded instruments exceeds the number of endogenous regressors. Amemiya [1], Hayashi [11], and Wooldridge [16] show that 2SLS provides an efficient way to exploit the available instruments under homoskedasticity. Section~3 derives the 2SLS estimator, relates it to IV in the just-identified case, and establishes its efficiency within the class of IV estimators under homoskedastic errors.

Generalized Method of Moments (GMM). GMM nests both IV and 2SLS as special cases. It begins from a set of moment conditions $\mathbb{E}[\mathbf{g}(\mathbf{w}_i, \boldsymbol{\beta}_0)] = \mathbf{0}$ and constructs the estimator by minimising a quadratic form in the sample analogue. With an *optimal* weighting matrix, GMM is asymptotically efficient within the class of GMM estimators based on the same moment conditions. As Hayashi [11], Hall [8], and Hansen [10] emphasize, GMM also extends naturally to heteroskedastic and autocorrelated settings by changing the weighting matrix and variance estimation rather than the underlying moment conditions. Section~4 provides a self-contained treatment.

Diagnostics. Identifying endogeneity in a given dataset, assessing instrument validity, and testing overidentifying restrictions are tasks of independent importance. Section~3.3 covers Hausman-type tests, weak-instrument diagnostics, the homoskedastic Sargan test, and the robust Hansen J-test.

2. Instrumental Variable Estimator

To address the endogeneity problem, we can introduce instrumental variables (IVs).

Definition 2.1 (Instrumental Variable)**.** A variable z_i is an *instrument* for the endogenous regressor x_i if it satisfies the following conditions:

(i) **Relevance:** $\text{Cov}(z_i, x_i) \neq 0$.

(ii) **Exogeneity:** $\mathbb{E}[z_i \varepsilon_i] = 0$.

The *relevance condition* ensures that the instrument is correlated with the endogenous regressor. This guarantees that the instrument contains identifying information about the endogenous variation in x_i and that the IV estimator is well-defined in the just-identified case. As Hayashi [11], Hansen [10], and Wooldridge [16] note, stronger relevance typically yields a more precise estimator, whereas weak relevance can lead to poor finite-sample performance and unreliable inference.

The *exogeneity condition* ensures that the instrument is not correlated with the error term, which is crucial for the consistency of the IV estimator. As Hayashi [11], Hansen [10], and Wooldridge [16] make clear, if this condition is violated, the IV estimator will be biased and inconsistent.

Before deriving the IV estimator, let us consider the following linear regression model with one endogenous regressor and two exogenous regressors:

$$y_i = \beta_1 + \beta_2 x_{2i} + \mathbf{x}_i^\top \boldsymbol{\gamma} + \varepsilon_i, \tag{3}$$

where x_{2i} is the endogenous regressor, $\mathbf{x}_i$ is a vector of exogenous regressors, and ε_i is the error term. We assume that we have an instrument z_{2i} for x_{2i} that satisfies the relevance and exogeneity conditions. In this specification, β_1 is the intercept, β_2 is the coefficient on the endogenous regressor x_{2i}, and $\boldsymbol{\gamma}$ is a vector of coefficients on the exogenous regressors.

We can also write the model in matrix form as follows:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{4}$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times k$ matrix of regressors (including the intercept, the endogenous regressor, and the exogenous regressors), $\boldsymbol{\beta}$ is the $k \times 1$ vector of parameters, and $\boldsymbol{\varepsilon}$ is the $n \times 1$ vector of error terms.

We can partition the matrix of regressors $\mathbf{X}$ into two parts: $\mathbf{X} = [\mathbf{X}_1 \,|\, \mathbf{X}_2]$, where $\mathbf{X}_1$ contains the exogenous regressors (including the intercept) and $\mathbf{X}_2$ contains the endogenous regressor x_2. We can define the vector of instruments $\mathbf{Z} = [\mathbf{X}_1 \,|\, \mathbf{z}_2]$, where $\mathbf{z}_2$ is the $n \times 1$ vector of the instrument for x_2. Since x_2 is endogenous, $\mathbb{E}[\mathbf{x}_i \varepsilon_i] \neq \mathbf{0}$. However, instrument exogeneity gives $\mathbb{E}[\mathbf{z}_i \varepsilon_i] = \mathbf{0}$. The IV estimator is derived by imposing the sample analogue of this population moment condition:

$$\frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}}) = \mathbf{0}. \tag{5}$$

We have substituted $\boldsymbol{\varepsilon}$ with $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}}$ in the moment condition.

Using the principle of the method of moments estimation, which states that the sample analogue of the population moment condition should hold at the estimated parameters, we can proceed with the derivation of the IV estimator, as follows:

$$\begin{aligned}
1/n[\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}})] &= \mathbf{0} \\
\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}}) &= \mathbf{0} \\
\mathbf{Z}'\mathbf{y} - \mathbf{Z}'\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}} &= \mathbf{0} \\
\mathbf{Z}'\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}} &= \mathbf{Z}'\mathbf{y} \\
\hat{\boldsymbol{\beta}}_{\text{IV}} &= (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}.
\end{aligned}$$

In the second line, we have multiplied both sides of the equation by n to eliminate the $1/n$ term. In the third line, we have rearranged the equation to isolate $\mathbf{Z}'\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{IV}}$ on one side. Finally, in the last line, we have solved for $\hat{\boldsymbol{\beta}}_{\text{IV}}$ by pre-multiplying both sides of the equation by $(\mathbf{Z}'\mathbf{X})^{-1}$.

Unlike OLS, which uses the orthogonality condition between regressors and errors, the IV estimator replaces that invalid condition with valid moment conditions involving the instruments $\mathbf{Z}$. In this sense, IV corrects the endogeneity problem by projecting identification onto exogenous variation in the instruments. The IV estimator is consistent for $\boldsymbol{\beta}$ when the instruments are valid, the model is correctly specified, and the relevant rank condition holds.

Proposition 2.2 (Consistency of the IV Estimator)**.** *Under the assumptions of the linear model, random sampling, full rank, and the validity of the instruments (i.e., relevance and exogeneity), the IV estimator* $\hat{\boldsymbol{\beta}}_{\text{IV}}$ *is consistent for* $\boldsymbol{\beta}$*.*

Proof. We can write the IV estimator as follows:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\mathrm{IV}} &= (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y} \\ &= (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta} + (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\boldsymbol{\varepsilon}.\end{aligned}$$

Since $n^{-1}\mathbf{Z}'\mathbf{X} \xrightarrow{p} \mathbb{E}[\mathbf{z}_i\mathbf{x}_i^\top]$, and the limit is non-singular by the relevance condition, the inverse exists with probability approaching one. Moreover, since $\mathbb{E}[\mathbf{z}_i\varepsilon_i] = \mathbf{0}$ by the exogeneity condition, we have $n^{-1}\mathbf{Z}'\boldsymbol{\varepsilon} \xrightarrow{p} \mathbf{0}$. Therefore, by Slutsky's theorem, $\hat{\boldsymbol{\beta}}_{\mathrm{IV}} \xrightarrow{p} \boldsymbol{\beta}$. $■$

Remark 2.3. The consistency of the IV estimator relies crucially on the validity of the instruments. Under conventional fixed-parameter asymptotics, nonzero but weak first-stage coefficients do not by themselves destroy consistency. They can nevertheless make IV estimates imprecise, biased in finite samples, and poorly approximated by conventional normal asymptotics. If relevance fails in the population, the parameter is not identified; if the instruments are correlated with the error term, the IV estimator is generally inconsistent. Therefore, it is essential to justify instrument validity and assess instrument strength in applied work.

Remark 2.4. Normally, we do not prove the unbiasedness of the IV estimator, because it is generally not unbiased in small samples and its finite-sample moments may not exist in some just-identified models. Under the stated regularity conditions it is consistent, meaning that it converges in probability to the true parameter value $\boldsymbol{\beta}$.

The IV idea extends naturally to models with multiple endogenous regressors and multiple instruments. In the just-identified multivariate case, where the total number of instruments equals the number of structural parameters, the estimator still takes the form $\hat{\boldsymbol{\beta}}_{\mathrm{IV}} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}$. If included exogenous regressors serve as their own instruments, this is equivalent to requiring the number of excluded instruments to equal the number of endogenous regressors. In the overidentified case, estimation is typically carried out using 2SLS or GMM rather than this simple just-identified formula. These issues are discussed in more detail in the subsequent sections on Two-Stage Least Squares (2SLS) and Generalized Method of Moments (GMM).

Remark 2.5. In the just-identified case, the IV estimator is equivalent to the Two-Stage Least Squares (2SLS) estimator. In the overidentified case, 2SLS provides a way to combine the available instruments.

3. Two-Stage Least Squares (2SLS) Estimator

The two-stage least squares (2SLS) estimator is a generalization of the IV estimator that allows for multiple endogenous regressors and multiple instruments. Wooldridge [16], Hayashi [11], and Amemiya [1] provide standard treatments of the 2SLS estimator, including its derivation, properties, and applications. The 2SLS estimator is particularly useful in the overidentified case, where the number of excluded instruments exceeds the number of endogenous regressors, as it allows for the efficient use of the available instruments under homoskedasticity. Following these treatments, we can specify the framework for deriving the 2SLS estimator as follows:

Consider a general linear regression model with k regressors. This model can be presented as follows:

$$y_i = \beta_1 + \beta_2 x_{2i} + \ldots + \beta_k x_{ki} + \varepsilon_i, \tag{6}$$

where y_i is the dependent variable for observation i and ε_i is the corresponding error term. The model includes k regressors, including the intercept. Let us assume that $x_2, \ldots, x_{k-1}$ are exogenous regressors, while x_k is an endogenous regressor. In this case, we have one endogenous regressor and $k-1$ exogenous regressors. This implies that $E[x_k \varepsilon_i] \neq 0$, while $E[x_j \varepsilon_i] = 0$ for $j = 2, \ldots, k-1$.

The m excluded instruments for the endogenous regressor x_k are denoted by $z_{1i}, \ldots, z_{mi}$. They satisfy the exogeneity conditions $E[z_{ji}\varepsilon_i] = 0$ for $j = 1, \ldots, m$. Relevance requires the excluded instruments to contribute sufficient predictive variation for the endogenous regressor conditional on the included exogenous regressors; individual pairwise covariances need not all be nonzero. The matrix of instruments can be denoted as $\mathbf{Z} = [\mathbf{X}_1 \mid \mathbf{Z}_2]$, where $\mathbf{X}_1$ contains the exogenous regressors (including the intercept) and $\mathbf{Z}_2$ contains the excluded instruments. The 2SLS estimator can be derived by first regressing the endogenous regressor x_k on the instruments $\mathbf{Z}$. This is the first stage of the 2SLS procedure, and it can be written as follows:

$$x_k = \mathbf{Z}\boldsymbol{\gamma} + v, \tag{7}$$

where $\boldsymbol{\gamma}$ is a vector of parameters to be estimated, and v is the error term in the first stage regression.

In equation (7), x_k is regressed on $\mathbf{Z}$, which includes the exogenous regressors and the excluded instruments. The OLS coefficient $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'x_k$ estimates the coefficients of the population linear projection of x_k on $\mathbf{Z}$. By the defining orthogonality property of a linear projection, its population residual satisfies $\mathbb{E}[\mathbf{z}_i v_i] = \mathbf{0}$. This statement does not follow merely from structural instrument exogeneity. The fitted values are $\hat{x}_k = \mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'x_k$. Using the fitted regressors, we can write the algebraic second-stage estimator for $\boldsymbol{\beta}$ as follows:

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y}, \tag{8}$$

Substitution of the expression for $\hat{\mathbf{X}}$ into the second stage regression, together with some algebraic manipulation, yields the 2SLS estimator in the following form:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\text{2SLS}} &= (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y}\\ &= (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y}\\ &= \left(\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}.\end{aligned}$$

where $\mathbf{P}_Z$ is the projection matrix onto the space spanned by the columns of $\mathbf{Z}$, defined as $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$. As Wooldridge [16], Hayashi [11], and Amemiya [1] explain, the 2SLS point estimator can be computed by replacing the endogenous regressors with their first-stage fitted values and running the second-stage regression. The standard errors must still use an IV/2SLS variance formula rather than naive OLS standard errors from that regression. Under the same conditions used for IV consistency, 2SLS is consistent for $\boldsymbol{\beta}$.

3.1 Variance-covariance Matrix of the 2SLS Estimator

3.1.1 Variance-covariance Matrix under Spherical Errors

As Hayashi [11], Amemiya [1], and Wooldridge [16] show, under homoskedasticity the conventional large-sample covariance estimator for 2SLS has the familiar projection-matrix form

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{\text{2SLS}}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1},$$

where $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$. Unlike OLS with exogenous regressors, 2SLS is generally not conditionally unbiased given $\mathbf{X}$ and $\mathbf{Z}$, so this expression should not be interpreted as an exact finite-sample conditional variance formula.

Here, σ^2 can be estimated using the structural residuals as follows:

$$\hat{\sigma}^2 = \frac{1}{n-k}\sum_{i=1}^{n}\hat{\varepsilon}_i^2,$$

where $\hat{\varepsilon}_i = y_i - \mathbf{x}_i^\top\hat{\boldsymbol{\beta}}_{\text{2SLS}}$ is the structural residual and k is the number of structural parameters estimated (including the intercept). The standard errors of the 2SLS estimator are then obtained from the square roots of the diagonal elements of $\hat{\sigma}^2(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}$.

3.1.2 Variance-covariance Matrix under Heteroskedasticity

In the presence of heteroskedasticity, the homoskedastic formula above is no longer valid. A standard heteroskedasticity-robust covariance estimator, discussed by White [15] and Hansen [10], is the sandwich form

$$\widehat{\text{Var}}_{\text{rob}}(\hat{\boldsymbol{\beta}}_{\text{2SLS}}) = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\hat{\boldsymbol{\Omega}}\,\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1},$$

where $\hat{\boldsymbol{\Omega}}$ is an estimator of the covariance matrix of the structural disturbances. In the simplest heteroskedastic but uncorrelated case, one uses $\hat{\boldsymbol{\Omega}} = \text{diag}(\hat{\varepsilon}_1^2, \ldots, \hat{\varepsilon}_n^2)$, where $\hat{\varepsilon}_i = y_i - \mathbf{x}_i^\top\hat{\boldsymbol{\beta}}_{\text{2SLS}}$ is the structural residual for observation i.

3.1.3 Variance-Covariance Matrix under Serial Correlation

With serial correlation, one replaces $\hat{\boldsymbol{\Omega}}$ in the robust sandwich formula by a heteroskedasticity-and-autocorrelation-consistent (HAC) estimator, such as the Newey–West estimator; see Hayashi [11] and Hansen [10]. The resulting covariance estimator is

$$\widehat{\mathrm{Var}}_{\mathrm{NW}}(\hat{\boldsymbol{\beta}}_{\mathrm{2SLS}}) = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\hat{\boldsymbol{\Omega}}_{\mathrm{NW}}\,\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1},$$

where $\hat{\boldsymbol{\Omega}}_{\mathrm{NW}}$ denotes a HAC estimator that incorporates both heteroskedasticity and autocorrelation up to a chosen lag length.

3.2 Properties of the 2SLS Estimator

3.2.1 Consistency

Proposition 3.1 (Consistency of the 2SLS Estimator)**.** *Under the assumptions of the linear model, random sampling, full rank, and the validity of the instruments, the 2SLS estimator $\hat{\boldsymbol{\beta}}_{\mathrm{2SLS}}$ is consistent for $\boldsymbol{\beta}$.*

Proof. The consistency of the 2SLS estimator can be established by showing that it converges in probability to $\boldsymbol{\beta}$ as the sample size increases. We can write the 2SLS estimator as follows:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\mathrm{2SLS}} &= \left(\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} \\ &= \boldsymbol{\beta} + \left(\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\boldsymbol{\varepsilon}.\end{aligned}$$

Since $\boldsymbol{\beta}$ is a constant vector, we can write its convergence to zero as follows:

$$\hat{\boldsymbol{\beta}}_{\mathrm{2SLS}} - \boldsymbol{\beta} = \left(\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\boldsymbol{\varepsilon}.$$

We can show that plim $\hat{\boldsymbol{\beta}}_{\mathrm{2SLS}} - \boldsymbol{\beta} = 0$ by showing that the right-hand side converges in probability to zero. We have $n^{-1}\mathbf{X}'\mathbf{Z} \xrightarrow{p} \mathbb{E}[\mathbf{x}_i\mathbf{z}_i^\top]$, $n^{-1}\mathbf{Z}'\mathbf{Z} \xrightarrow{p} \mathbb{E}[\mathbf{z}_i\mathbf{z}_i^\top]$, and $n^{-1}\mathbf{Z}'\boldsymbol{\varepsilon} \xrightarrow{p} \mathbb{E}[\mathbf{z}_i\varepsilon_i] = \mathbf{0}$, where the last equality follows from the exogeneity condition of the instruments. Therefore, by Slutsky's theorem, $\hat{\boldsymbol{\beta}}_{\mathrm{2SLS}} \xrightarrow{p} \boldsymbol{\beta}$. \$■\$

Remark 3.2. As Wooldridge [16], Hayashi [11], and Hansen [10] emphasize, the consistency of the 2SLS estimator relies on the same conditions as the IV estimator, including instrument validity and the population rank condition. Weak instruments can produce imprecise estimates, substantial finite-sample bias, and unreliable conventional inference even when the standard fixed-parameter consistency result applies. If the instruments are correlated with the error term, the 2SLS estimator is generally inconsistent. Therefore, it is essential to justify instrument validity and assess instrument strength in applied work.

Remark 3.3. As Amemiya [1], Hayashi [11], and Wooldridge [16] note, in the just-identified case, the 2SLS estimator is equivalent to the IV estimator. In the overidentified case, 2SLS combines the available instruments and is efficient within the relevant class of IV estimators under homoskedasticity.

3.2.2 Asymptotic Normality

The asymptotic normality of the OLS estimator can be derived from the following theorem; see Hayashi [11], Wooldridge [16], and Hansen [10].

Theorem 3.4 (Asymptotic Normality of the OLS Estimator)**.** *Under the assumptions of the linear model, random sampling, full rank, strict exogeneity, and homoskedasticity, the OLS estimator* $\hat{\boldsymbol{\beta}}_{\text{OLS}}$ *is asymptotically normal, i.e.,*

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \sigma^2(\mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top])^{-1}), \tag{9}$$

where σ^2 *is the variance of the error term* ε_i*.*

As Amemiya [1], Hayashi [11], and Wooldridge [16] show, the asymptotic normality of the 2SLS estimator can be established under conditions analogous to those for OLS, together with instrument validity and sufficient strength. The asymptotic distribution of the 2SLS estimator is given by:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{2SLS}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{2SLS}}), \tag{10}$$

where, under homoskedasticity,

$$\boldsymbol{\Sigma}_{\text{2SLS}} = \sigma^2 \left[\mathbb{E}(\mathbf{x}_i\mathbf{z}_i^\top)\, \mathbb{E}(\mathbf{z}_i\mathbf{z}_i^\top)^{-1}\, \mathbb{E}(\mathbf{z}_i\mathbf{x}_i^\top)\right]^{-1}.$$

Under heteroskedasticity or serial correlation, the appropriate sandwich covariance matrix replaces this simplified expression.

As Hayashi [11], Hansen [10], and Wooldridge [16] explain, asymptotic normality allows one to construct confidence intervals and perform hypothesis tests for the parameters of interest, using the appropriate IV/2SLS variance formula rather than the OLS one.

Remark 3.5. The conventional normal approximation for 2SLS relies on valid, sufficiently strong instruments. If instruments are weak, conventional confidence intervals and tests may be unreliable. Weak-identification-robust procedures, such as Anderson–Rubin inference, may then be appropriate. Stock–Yogo critical values can help diagnose weak identification in the settings for which they were derived, but they are not themselves an alternative inferential procedure.

3.2.3 Efficiency

As Amemiya [1], Hayashi [11], and Wooldridge [16] show, under homoskedasticity 2SLS is efficient within the class of IV estimators based on the same valid instrument set. Relative to an estimator that uses only a subset of valid instruments, using additional relevant instruments can improve asymptotic precision.

The efficiency gain from combining valid instruments can be substantial, especially when the additional instruments strongly predict the endogenous regressor. Invalid instruments generally destroy consistency. Weak instruments create a different problem: they can make the finite-sample performance of 2SLS poor and conventional asymptotic inference unreliable. Amemiya [1], Hayashi [11], and Wooldridge [16] therefore emphasize careful consideration of the choice of instruments in practice.

3.3 Testing for Endogeneity

When valid instruments are available, it can be useful to test whether the regressors treated as endogenous could instead be treated as exogenous. Under the null, OLS is consistent and typically more efficient than 2SLS; under the alternative, OLS is inconsistent while IV/2SLS remains consistent if the instruments are valid. As Hansen [10] and Wooldridge [16] note, one common approach is a Hausman-type test.

As Hansen [10] and Wooldridge [16] explain, this class of tests compares the OLS and 2SLS estimators to determine whether there is a systematic difference between them. In the classical homoskedastic setting, a standard Hausman statistic is

$$H = (\hat{\boldsymbol{\beta}}_{\text{2SLS}} - \hat{\boldsymbol{\beta}}_{\text{OLS}})^\top \left(\text{Var}(\hat{\boldsymbol{\beta}}_{\text{2SLS}}) - \text{Var}(\hat{\boldsymbol{\beta}}_{\text{OLS}})\right)^{-1} (\hat{\boldsymbol{\beta}}_{\text{2SLS}} - \hat{\boldsymbol{\beta}}_{\text{OLS}}),$$

where $\text{Var}(\hat{\boldsymbol{\beta}}_{\text{OLS}})$ and $\text{Var}(\hat{\boldsymbol{\beta}}_{\text{2SLS}})$ are the asymptotic variances of the OLS and 2SLS estimators, respectively. Under standard regularity conditions, this statistic is asymptotically chi-squared with degrees of freedom equal to the number of tested coefficients. In practice, robust endogeneity diagnostics are often implemented as Durbin–Wu–Hausman regression-based tests, which remain valid under heteroskedasticity when paired with robust covariance estimation.

As Hansen [10] and Wooldridge [16] caution, Hausman-type tests are not definitive tests for endogeneity. Their interpretation relies on maintained assumptions, especially the validity of the instruments used by 2SLS. These diagnostics are therefore best used alongside broader specification and instrument-validity checks.

3.4 Weak Instruments

Weak instruments arise when the instruments only weakly predict the endogenous regressors after conditioning on the included exogenous variables. This is not itself a form of endogeneity; rather, it is weak identification. Weak instruments can make 2SLS estimates imprecise and can also induce substantial finite-sample bias, causing IV estimates to drift toward OLS. As Hansen [10] and Chaussé [5] discuss, Stock and Yogo (2005) developed critical values for particular weak-identification diagnostics.

In the homoskedastic linear IV setting, Stock–Yogo critical values are tabulated for the Cragg–Donald statistic; with one endogenous regressor this is closely related to the conventional first-stage excluded-instrument F-statistic. These critical values are not automatically valid for heteroskedasticity-robust Kleibergen–Paap statistics. When instruments may be weak, researchers should report appropriate diagnostics and consider estimators such as LIML together with weak-identification-robust inference such as Anderson–Rubin tests.

3.5 Overidentification, the Sargan Test, and the Hansen J-Test

When the number of moment conditions exceeds the number of structural parameters, the model is overidentified. When included exogenous regressors instrument themselves, this is equivalent to having more excluded instruments than endogenous regressors. As Hayashi [11], Hall [8], and Hansen [10] explain, overidentification permits a test of whether the maintained moment conditions are jointly compatible with the data. The Hansen J-statistic is given by:

$$J = n\, g_n(\hat{\boldsymbol{\beta}})' \hat{\mathbf{W}}\, g_n(\hat{\boldsymbol{\beta}}),$$

where $g_n(\hat{\boldsymbol{\beta}}) = n^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$ is the vector of sample moment conditions evaluated at the efficient GMM estimator and $\hat{\mathbf{W}}$ consistently estimates the inverse asymptotic covariance matrix of the moments. In the homoskedastic linear IV case, the corresponding overidentification test is the familiar Sargan test based on the IV residuals.

As Hayashi [11], Hall [8], and Hansen [10] show, under the null hypothesis that all overidentifying restrictions are valid and under the relevant regularity conditions, the Hansen J-statistic is asymptotically chi-squared with degrees of freedom equal to the number of overidentifying restrictions, that is, the number of moment conditions minus the number of estimated parameters. If the test statistic exceeds the relevant critical value, the null is rejected, suggesting that at least one instrument or maintained moment restriction is invalid.

4. Generalized Method of Moments (GMM) Estimator

The generalized method of moments (GMM) estimator generalizes the instrumental variables (IV) estimator by using moment conditions implied by the model's assumptions. Like IV and 2SLS, GMM relies on valid instruments and can accommodate multiple endogenous regressors, multiple instruments, and both just-identified and overidentified models. As Hayashi [11], Hall [8], and Hansen [10] explain, its broader appeal lies in its flexibility: GMM can be applied to linear and nonlinear models, allows for homoskedastic or heteroskedastic errors, and, with the optimal weighting matrix, efficiently combines multiple moment conditions to estimate the parameters of interest.

Given the following linear regression model (in matrix form):

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{11}$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times k$ matrix of regressors (including the intercept, the endogenous regressor, and the exogenous regressors), $\boldsymbol{\beta}$ is the $k \times 1$ vector of parameters, and $\boldsymbol{\varepsilon}$ is the $n \times 1$ vector of error terms. Let $\mathbf{Z}$ be the $n \times m$ matrix of instruments, where $m \geq k$, and assume that the following moment conditions hold:

$$E[\mathbf{z}_i(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})] = \mathbf{0}. \tag{12}$$

As Hayashi [11], Hall [8], and Hansen [10] define it, the GMM estimator $\hat{\boldsymbol{\beta}}_{\text{GMM}}$ is the value of $\boldsymbol{\beta}$ that minimizes the quadratic form of the sample moment conditions, weighted by a positive definite weighting matrix $\mathbf{W}_n$:

$$\hat{\boldsymbol{\beta}}_{\text{GMM}} = \arg\min_{\boldsymbol{\beta}} \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{z}_i(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}) \right)^\top \mathbf{W}_n \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{z}_i(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}) \right). \tag{13}$$

In matrix notation, the objective function can be written as:

$$Q_n(\boldsymbol{\beta}) = \left(\frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right)^\top \mathbf{W}_n \left(\frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right). \tag{14}$$

For a symmetric weighting matrix, the GMM estimator is obtained by minimizing the objective function $Q_n(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$, and setting the derivative equal to zero as follows:

$$\frac{\partial Q_n(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2 \left(\frac{1}{n} \mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right)^\top \mathbf{W}_n \left(\frac{1}{n} \mathbf{Z}'\mathbf{X} \right) = \mathbf{0}. \tag{15}$$

From which the GMM estimator can be derived, step by step, as follows:

$$
\begin{aligned}
\left(\frac{1}{n}\mathbf{Z}'(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})\right)^{\top}\mathbf{W}_n\left(\frac{1}{n}\mathbf{Z}'\mathbf{X}\right) &= \mathbf{0}\\
\left(\frac{1}{n}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'\mathbf{Z}\right)\mathbf{W}_n\left(\frac{1}{n}\mathbf{Z}'\mathbf{X}\right) &= \mathbf{0}\\
(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'\mathbf{Z}\mathbf{W}_n(\mathbf{Z}'\mathbf{X}) &= \mathbf{0}\\
(\mathbf{y}'-\boldsymbol{\beta}'\mathbf{X}')\mathbf{Z}\mathbf{W}_n(\mathbf{Z}'\mathbf{X}) &= \mathbf{0}\\
\mathbf{y}'\mathbf{Z}\mathbf{W}_n(\mathbf{Z}'\mathbf{X})-\boldsymbol{\beta}'\mathbf{X}'\mathbf{Z}\mathbf{W}_n(\mathbf{Z}'\mathbf{X}) &= \mathbf{0}\\
\boldsymbol{\beta}'\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X} &= \mathbf{y}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}\\
(\boldsymbol{\beta}'\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X})' &= (\mathbf{y}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X})'\\
(\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X})'\boldsymbol{\beta} &= (\mathbf{Z}'\mathbf{X})'\mathbf{W}_n'\mathbf{Z}'\mathbf{y}\\
\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}\,\boldsymbol{\beta} &= \mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{y}\\
\boldsymbol{\beta} &= (\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{y}
\end{aligned}
$$

In the second step, we used the fact that the transpose of a product of matrices is the product of their transposes in reverse order. In the third step, we multiplied both sides by the nonzero scalar n^2 to remove the scale factors. We then rearranged terms, transposed both sides, used the symmetry of $\mathbf{W}_n$, and solved for $\boldsymbol{\beta}$ by pre-multiplying both sides by the inverse of $\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}$.

From this derivation, we can see that the GMM estimator can be expressed as:

$$
\hat{\boldsymbol{\beta}}_{\text{GMM}} = (\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X})^{-1}\,\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{y}.
$$

4.1 The Weighting Matrix $\mathbf{W}_n$

As Hall [8], Hayashi [11], and Hansen [10] explain, the role of the weighting matrix $\mathbf{W}_n$ in the GMM objective function is to account for the variance-covariance structure of the moment conditions, weighting them inversely to their variance and accounting for potential correlation. In the first step of GMM estimation, any matrix that satisfies the following conditions can be used as the weighting matrix $\mathbf{W}_n$:

(1) it must be positive definite, which ensures that the quadratic form is nonnegative; identification is additionally required for a unique population minimum;
(2) it should converge in probability to a positive definite matrix $\mathbf{W}$ as the sample size increases, which contributes to consistency together with correct specification and identification; and
(3) its dimension must be $m \times m$ (where m is the number of moment conditions), which ensures that the objective function is properly defined.

In most applications, in the first step of GMM estimation, a simple choice for the weighting matrix is the identity matrix $\mathbf{W}_n = \mathbf{I}_m$, which gives equal weight to all moment conditions. While this choice yields a consistent estimator, it is generally not asymptotically efficient, as it does not account for the variance-covariance structure of the moment conditions (which may exhibit heteroskedasticity or correlation).

In practice, many matrices satisfy the conditions for $\mathbf{W}_n$, which implies there are many GMM estimators that are consistent for $\boldsymbol{\beta}$. However, these estimators have different asymptotic efficiency properties. As Hayashi [11], Hall [8], and Hansen [10] show, the optimal weighting matrix is $\mathbf{W}_n = \mathbf{S}_n^{-1}$, where $\mathbf{S}_n$ is a consistent estimator of the variance-covariance matrix of the moment conditions. This optimal choice produces the asymptotically efficient GMM estimator, which achieves the lowest asymptotic variance among all GMM estimators based on the same set of moment conditions.

4.2 The Optimal Weighting Matrix

Definition 4.1 (Optimal Weighting Matrix)**.** In the linear i.i.d. GMM setting, the optimal weighting matrix is the inverse of the asymptotic covariance matrix of the sample moments. A common heteroskedasticity-robust estimator takes the form $\mathbf{W}_n = \mathbf{S}_n^{-1}$, where $\mathbf{S}_n$ is a consistent estimator of that covariance matrix, for example

$$\mathbf{S}_n = \frac{1}{n}\sum_{i=1}^{n} \mathbf{z}_i \mathbf{z}_i^\top (y_i - \mathbf{x}_i^\top \tilde{\boldsymbol{\beta}})^2,$$

where $\tilde{\boldsymbol{\beta}}$ is an initial consistent estimator. With serial dependence, $\mathbf{S}_n$ should instead be replaced by a consistent long-run covariance estimator.

Proposition 4.2 (Efficiency of the Optimal GMM Estimator)**.** *The GMM estimator that uses the optimal weighting matrix $\mathbf{W}_n = \mathbf{S}_n^{-1}$ is asymptotically efficient among the class of GMM estimators that use the same set of moment conditions. As Hayashi [11], Hall [8], and Hansen [10] show, this means that the asymptotic variance of the optimal GMM estimator is less than or equal to the asymptotic variance of any other GMM estimator that uses a different weighting matrix, in the positive semi-definite sense.*

Proof. Let $\mathbf{G}$ denote the Jacobian of the population moments and let Ω denote the asymptotic covariance matrix of $\sqrt{n}\, g_n(\beta_0)$. For a generic limiting weighting matrix $\mathbf{W}$, the asymptotic variance of the GMM estimator is

$$(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}\, \mathbf{G}'\mathbf{W}\Omega\mathbf{W}\mathbf{G}\, (\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}\,.$$

When $\mathbf{W} = \Omega^{-1}$, this simplifies to

$$\left(\mathbf{G}'\Omega^{-1}\mathbf{G}\right)^{-1},$$

which is the smallest variance in the positive semi-definite ordering among GMM estimators based on the same moment conditions. Hence the optimal-weight GMM estimator is asymptotically efficient within that class. $■$

Remark 4.3. The optimal GMM estimator is particularly useful in the presence of heteroskedasticity or autocorrelation in the error terms, as it accounts for the variance-covariance structure of the moment conditions, leading to more efficient estimates. However, it is important to note that the optimal GMM estimator relies on a consistent estimator of the variance-covariance matrix of the moment conditions, which can be challenging to obtain in practice, especially in the presence of heteroskedasticity or autocorrelation. In such cases, alternative methods for estimating the variance-covariance matrix, such as heteroskedasticity-robust or Newey–West estimators, may be necessary to ensure the efficiency of the GMM estimator.

4.3 Second Step GMM Estimation

In practice, the optimal GMM estimator is often implemented using a two-step procedure. As Hall [8], Hayashi [11], Hansen [10], and Chaussé [4] explain, the first step uses a consistent but potentially inefficient weighting matrix, such as the identity matrix $\mathbf{W}_n = \mathbf{I}_m$, to obtain an initial estimator and construct a consistent estimator of the variance-covariance matrix of the moment conditions, $\mathbf{S}_n$. In the second step, the optimal weighting matrix $\mathbf{W}_n = \mathbf{S}_n^{-1}$ is used to re-estimate the parameters, yielding the efficient GMM estimator. The two-step GMM estimator can be expressed as follows:

$$\hat{\boldsymbol{\beta}}^{(2)}_{\text{GMM}} = \left(\mathbf{X}'\mathbf{Z}\mathbf{S}_n^{-1}\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}\mathbf{S}_n^{-1}\mathbf{Z}'\mathbf{y}, \tag{16}$$

where $\mathbf{S}_n$ is the consistent estimator of the variance-covariance matrix of the moment conditions obtained from the first step. The two-step GMM estimator is consistent and asymptotically efficient under the same conditions as the optimal GMM estimator, and it is widely used in practice due to its computational convenience and efficiency properties.

$\mathbf{S}_n$ is estimated using the residuals from the first-step GMM estimation as follows:

$$\mathbf{S}_n = \frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\mathbf{z}_i^\top(y_i - \mathbf{x}_i^\top\hat{\boldsymbol{\beta}}^{(1)}_{\text{GMM}})^2,$$

where $\hat{\boldsymbol{\beta}}^{(1)}_{\text{GMM}}$ is the first-step GMM estimator obtained using a simple weighting matrix, such as the identity matrix $\mathbf{W}_n = \mathbf{I}_m$. The second-step GMM estimator $\hat{\boldsymbol{\beta}}^{(2)}_{\text{GMM}}$ is then obtained by using the optimal weighting matrix $\mathbf{W}_n = \mathbf{S}_n^{-1}$ in the GMM estimation, which yields the efficient GMM estimator. The optimal weighting matrix $\mathbf{W}_n$ can be computed using the formula:

$$\mathbf{W}_n = \mathbf{S}_n^{-1} = \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{z}_i \mathbf{z}_i^\top (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_{\text{GMM}}^{(1)})^2 \right)^{-1}.$$

4.4 Properties of the GMM Estimators

4.4.1 Consistency

As Hayashi [11], Hall [8], and Hansen [10] explain, the GMM estimator is consistent under the following conditions:

1. The population moment condition is correctly specified:

$$E[\mathbf{z}_i u_i(\beta_0)] = \mathbf{0}, \qquad u_i(\beta_0) = y_i - \mathbf{x}_i'\beta_0.$$

2. The sample moments satisfy a uniform law of large numbers, so that

$$g_n(\beta) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{z}_i \{y_i - \mathbf{x}_i'\beta\}$$

   converges uniformly in probability to

$$g(\beta) = E[\mathbf{z}_i \{y_i - \mathbf{x}_i'\beta\}].$$

3. The parameter space is compact, or the estimator lies in a compact set with probability approaching one.

4. The weighting matrix satisfies

$$\mathbf{W}_n \xrightarrow{p} \mathbf{W},$$

   where $\mathbf{W}$ is symmetric positive definite.

5. The true parameter β_0 is identified, meaning that

$$g(\beta) = \mathbf{0} \quad \text{if and only if} \quad \beta = \beta_0.$$

   In the linear IV/GMM model, this requires an appropriate rank condition, for example

$$\text{rank}\{E(\mathbf{z}_i \mathbf{x}_i')\} = k,$$

   where k is the number of parameters.

Proposition 4.4 (Consistency of the Linear GMM Estimator)**.** *Let*

$$g_n(\beta) = \frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\beta)$$

and define the GMM estimator by

$$\hat{\beta}_{\text{GMM}} = \arg\min_{\beta\in\mathcal{B}} Q_n(\beta), \qquad Q_n(\beta) = g_n(\beta)'\mathbf{W}_n g_n(\beta).$$

Suppose that $g_n(\beta)$ converges uniformly in probability to $g(\beta)$, that $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$, where $\mathbf{W}$ is positive definite, and that the population criterion

$$Q(\beta) = g(\beta)'\mathbf{W}g(\beta)$$

has a unique minimum at β_0. Then

$$\hat{\beta}_{\text{GMM}} \xrightarrow{p} \beta_0.$$

Proof. By the uniform law of large numbers,

$$\sup_{\beta\in\mathcal{B}} \|g_n(\beta) - g(\beta)\| \xrightarrow{p} 0.$$

Since $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$, it follows that

$$\sup_{\beta\in\mathcal{B}} |Q_n(\beta) - Q(\beta)| \xrightarrow{p} 0,$$

where

$$Q(\beta) = g(\beta)'\mathbf{W}g(\beta).$$

By correct specification,

$$g(\beta_0) = \mathbf{0},$$

so that

$$Q(\beta_0) = 0.$$

Because $\mathbf{W}$ is positive definite and the parameter is identified,

$$Q(\beta) > 0 \qquad \text{for all } \beta \neq \beta_0.$$

Thus $Q(\beta)$ has a unique minimum at β_0. Uniform convergence of Q_n to Q, together with uniqueness of the population minimizer, implies by the extremum estimator consistency theorem that

$$\hat{\beta}_{\text{GMM}} \xrightarrow{p} \beta_0.$$

\$■\$

As Hayashi [11], Hall [8], and Hansen [10] note, in the linear IV/GMM model one can also prove consistency directly from the closed-form estimator rather than only through the general extremum-estimator argument. This approach exploits the linear structure of the model and the probability limits of the sample moment matrices. The GMM estimator can be written as follows:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\text{GMM}} &= \left(\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{y} \\ &= \boldsymbol{\beta}_0 + \left(\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\boldsymbol{\varepsilon}.\end{aligned}$$

Since $\boldsymbol{\beta}_0$ is a constant vector, we can write its convergence to zero as follows:

$$\hat{\boldsymbol{\beta}}_{\text{GMM}} - \boldsymbol{\beta}_0 = \left(\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\boldsymbol{\varepsilon}.$$

We can show that $\operatorname{plim}(\hat{\boldsymbol{\beta}}_{\text{GMM}} - \boldsymbol{\beta}_0) = 0$ by showing that the right-hand side converges in probability to zero. We have $n^{-1}\mathbf{X}'\mathbf{Z} \xrightarrow{p} \mathbb{E}[\mathbf{x}_i\mathbf{z}_i^\top]$, $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$, and $n^{-1}\mathbf{Z}'\boldsymbol{\varepsilon} \xrightarrow{p} \mathbb{E}[\mathbf{z}_i\varepsilon_i] = \mathbf{0}$, where the last equality follows from the exogeneity condition of the instruments. Therefore,

$$\begin{aligned}\left(\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\boldsymbol{\varepsilon} &= \left(\frac{1}{n}\mathbf{X}'\mathbf{Z}\,\mathbf{W}_n\,\frac{1}{n}\mathbf{Z}'\mathbf{X}\right)^{-1}\left(\frac{1}{n}\mathbf{X}'\mathbf{Z}\,\mathbf{W}_n\,\frac{1}{n}\mathbf{Z}'\boldsymbol{\varepsilon}\right) \\ &\xrightarrow{p} \left(\mathbb{E}[\mathbf{x}_i\mathbf{z}_i^\top]\mathbf{W}\,\mathbb{E}[\mathbf{z}_i\mathbf{x}_i^\top]\right)^{-1}\left(\mathbb{E}[\mathbf{x}_i\mathbf{z}_i^\top]\mathbf{W}\mathbf{0}\right) \\ &= \mathbf{0}.\end{aligned}$$

Thus, we have shown that $\operatorname{plim}\hat{\boldsymbol{\beta}}_{\text{GMM}} - \boldsymbol{\beta}_0 = 0$, which implies that $\hat{\boldsymbol{\beta}}_{\text{GMM}}$ is consistent for $\boldsymbol{\beta}_0$.

4.4.2 Asymptotic Normality

As Hayashi [11], Hall [8], and Hansen [10] show, the GMM estimator is asymptotically normal under the following regularity conditions:

(i) The GMM estimator $\hat{\beta}_{\text{GMM}}$ is consistent for the true parameter vector β_0.

(ii) The moment conditions are correctly specified:

$$E[\mathbf{z}_i u_i(\beta_0)] = \mathbf{0}.$$

(iii) The sample moments satisfy a central limit theorem:

$$\sqrt{n}g_n(\beta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Omega),$$

where

$$g_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{z}_i u_i(\beta),$$

and Ω is the long-run variance matrix of the moment conditions.

(iv) The weighting matrix satisfies

$$\mathbf{W}_n \xrightarrow{p} \mathbf{W},$$

where $\mathbf{W}$ is symmetric positive definite.

(v) The population moment function $g(\beta) = E[g_i(\beta)]$ is continuously differentiable in a neighborhood of β_0, with Jacobian

$$\mathbf{G} = \left.\frac{\partial g(\beta)}{\partial \beta'}\right|_{\beta=\beta_0}.$$

(vi) The rank condition holds:

$$\operatorname{rank}(\mathbf{G}) = k,$$

where k is the number of parameters.

Under these conditions,

$$\sqrt{n}\left(\hat{\beta}_{\text{GMM}} - \beta_0\right) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \Sigma_{\text{GMM}}\right),$$

where

$$\Sigma_{\text{GMM}} = (\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}\mathbf{G}'\mathbf{W}\Omega\mathbf{W}\mathbf{G}(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}.$$

For the linear IV/GMM model,

$$u_i(\beta) = y_i - \mathbf{x}_i'\beta,$$

so that

$$\mathbf{G} = E\left[\frac{\partial}{\partial \beta'}\mathbf{z}_i(y_i - \mathbf{x}_i'\beta)\right] = -E(\mathbf{z}_i\mathbf{x}_i').$$

Since the sign disappears in the sandwich variance formula, the asymptotic variance can equivalently be written using

$$\mathbf{Q}_{ZX} = E(\mathbf{z}_i\mathbf{x}_i').$$

Thus,

$$\Sigma_{\text{GMM}} = (\mathbf{Q}_{XZ}\mathbf{W}\mathbf{Q}_{ZX})^{-1}\mathbf{Q}_{XZ}\mathbf{W}\Omega\mathbf{W}\mathbf{Q}_{ZX}(\mathbf{Q}_{XZ}\mathbf{W}\mathbf{Q}_{ZX})^{-1},$$

where

$$\mathbf{Q}_{XZ} = E(\mathbf{x}_i\mathbf{z}_i') \quad \text{and} \quad \mathbf{Q}_{ZX} = E(\mathbf{z}_i\mathbf{x}_i').$$

This asymptotic normality result provides the basis for large-sample inference in the linear IV/GMM model. It allows one to construct confidence intervals and conduct hypothesis tests for the parameters of interest using the limiting distribution of the GMM estimator. As Hayashi [11], Hall [8], and Hansen [10] explain, the choice of weighting matrix affects the asymptotic variance of the estimator. The efficient GMM estimator is obtained when the limiting weighting matrix is chosen as $\mathbf{W} = \Omega^{-1}$, where Ω is the asymptotic variance of the moment conditions. In practice, Ω is replaced by a consistent estimator, often constructed from the sample analog of the moment covariance matrix, with heteroskedasticity- or autocorrelation-robust versions used when appropriate.

4.4.3 Efficiency

As Hayashi [11], Hall [8], and Hansen [10] show, among GMM estimators based on the same set of moment conditions, the asymptotically efficient estimator is obtained by choosing the limiting weighting matrix to be the inverse of the asymptotic variance-covariance matrix of the sample moments. Thus, the efficient weighting matrix is $\mathbf{W} = \Omega^{-1}$, where Ω denotes the asymptotic variance of $\sqrt{n}g_n(\beta_0)$. Under this optimal weighting matrix, the general GMM asymptotic variance formula reduces to

$$\Sigma_{\text{GMM}} = (\mathbf{G}'\Omega^{-1}\mathbf{G})^{-1}.$$

This is the smallest asymptotic variance attainable within the class of GMM estimators using the same moment conditions.

4.5 Tests

4.5.1 Overidentifying restrictions

The GMM framework provides several specification tests, including tests of overidentifying restrictions, tests of parameter restrictions, and more general tests of model specification. As Hall [8], Hansen [10], and Hayashi [11] explain, one of the most widely used specification tests in GMM is the Hansen J-test. In linear IV models, the homoskedastic special case is commonly called the Sargan test. The test examines whether the sample moment conditions are sufficiently close to zero when evaluated at the GMM estimator.

Let

$$g_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} g_i(\beta)$$

denote the vector of sample moments, and let $\hat{\beta}_{\text{GMM}}$ be the GMM estimator. The J-statistic is given by

$$J = n\, g_n(\hat{\beta}_{\text{GMM}})' \mathbf{W}_n g_n(\hat{\beta}_{\text{GMM}}),$$

where $\mathbf{W}_n$ is the weighting matrix used in the final-stage GMM criterion.

Under the null hypothesis that the overidentifying restrictions are valid, and provided that $\mathbf{W}_n$ consistently estimates the optimal weighting matrix Ω^{-1}, the statistic has the limiting distribution

$$J \xrightarrow{d} \chi^2_{q-k},$$

where q is the number of moment conditions and k is the number of estimated parameters. Thus, $q - k$ is the number of overidentifying restrictions. In the linear IV model, this is often interpreted as the number of instruments excluded from the structural equation in excess of the number of endogenous regressors, or more generally as the number of instruments minus the number of parameters estimated.

As Hall [8], Hansen [10], and Hayashi [11] explain, if the value of J exceeds the relevant critical value from the χ^2_{q-k} distribution, the null hypothesis is rejected. Rejection suggests that the moment conditions are not jointly compatible with the data and the maintained model. This may indicate that one or more instruments are invalid, but it may also reflect other forms of model misspecification, such as an incorrect functional form, omitted variables, or violations of the maintained error structure. The Hansen J-test is therefore an important diagnostic tool in applied GMM estimation, especially in overidentified models.

4.6 Other Diagnostic Tests

As Hall [8], Hansen [10], Chaussé [5], and Chaussé [4] emphasize, besides the test of overidentifying restrictions, several diagnostic tests are important in IV/GMM estimation. These include tests of endogeneity, such as the Durbin–Wu–Hausman test; tests of instrument relevance and weak identification, such as the first-stage F-statistic, the Cragg–Donald statistic, and the Kleibergen–Paap rank tests; tests of underidentification; and Wald tests of parameter restrictions. In dynamic panel GMM, additional diagnostics are essential, including Arellano–Bond tests for serial correlation and Difference-in-Hansen tests for subsets of instruments. Researchers should also examine heteroskedasticity, serial correlation, robustness across alternative instrument sets, and the possibility of instrument proliferation. Together, these tests help assess whether the model is identified, whether the instruments are valid and relevant, and whether inference based on the GMM estimator is reliable.

5. Synthesis and Summary

This handout has examined the econometric problem of endogeneity and the main estimation strategies used to obtain consistent estimates when ordinary least squares is no longer reliable. The starting point is the classical linear model, in which OLS is consistent and asymptotically normal when the regressors are exogenous, that is, when the regressors are uncorrelated with the structural disturbance. Once this condition fails, OLS converges to the wrong probability limit, and the resulting inconsistency cannot be solved simply by increasing the sample size. Endogeneity is therefore an identification problem rather than merely a sampling problem.

The handout has emphasized three major sources of endogeneity: omitted variables, simultaneity, and measurement error. In each case, the core issue is that one or more regressors contain variation that is systematically related to the error term. For example, omitted ability in a returns-to-schooling equation may make schooling endogenous; simultaneous determination of price and quantity may make price endogenous in demand or supply equations; and classical measurement error may mechanically induce correlation between the observed regressor and the composite error term. These examples show that endogeneity is not a technical nuisance but a central threat to causal interpretation in applied economics.

Instrumental variables provide a way to restore identification by replacing the invalid orthogonality condition between regressors and errors with valid moment conditions involving instruments. A valid instrument must be relevant, meaning that it provides predictive variation for the endogenous regressors conditional on the included exogenous variables, and exogenous, meaning that it is uncorrelated with the structural error term. In the just-identified case, the IV estimator is obtained by solving the sample analogue of the instrument-error orthogonality condition. In overidentified models, where the number of moment conditions exceeds the number of structural parameters, two-stage least squares provides a systematic way to combine the available instruments by projecting the endogenous regressors onto the instrument space and then estimating the structural equation using the fitted values.

The generalized method of moments extends these ideas by treating IV and 2SLS as special cases of estimation based on moment conditions. GMM estimates the parameter vector by minimizing a quadratic form in the sample moments, with the weighting matrix determining how different moment conditions are combined. Any positive definite weighting matrix satisfying suitable regularity conditions can yield a consistent estimator, but the efficient GMM estimator is obtained by weighting the moments by the inverse of their asymptotic variance-covariance matrix. The handout also stresses that a credible empirical analysis should report diagnostics for endogeneity, weak instruments, underidentification, overidentifying restrictions, parameter restrictions, heteroskedasticity, serial correlation, and, in dynamic panel settings, serial correlation and instrument proliferation. These tools help assess the model, but no collection of diagnostics can by itself prove instrument validity or establish a causal interpretation.

Table~1 summarizes the relationship among the simple instrumental variables estimator, the two-stage least squares estimator, and the generalized method of moments estimator. The three estimators share the same basic objective: to obtain consistent estimates when one or more regressors are endogenous by replacing invalid regressor-error orthogonality conditions with valid instrument-based moment conditions. The simple IV estimator is most transparent in the just-identified case, 2SLS extends the IV logic to settings with multiple instruments by projecting endogenous regressors onto the instrument space, and GMM provides the most general framework by estimating parameters from moment conditions using a weighting matrix. The table also highlights that the estimators differ mainly in how they use instruments and moment conditions, how they combine available information, and the assumptions required for consistency and efficient inference.

Table 1: Summary of the simple IV, 2SLS, and GMM estimators

Estimator	**Core idea**	**Main formula**	**Key conditions and interpretation**
Simple IV estimator	Uses an external source of exogenous variation to isolate variation in the endogenous regressor that is uncorrelated with the structural error term. It is most transparent in the just-identified case, where the total number of instruments equals the number of structural parameters.	$\hat{\boldsymbol{\beta}}_{\text{IV}} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}$ when $\mathbf{Z}'\mathbf{X}$ is square and nonsingular.	Requires instrument relevance and exogeneity: $\text{rank}\{E(\mathbf{z}_i\mathbf{x}_i')\} = k$ and $E(\mathbf{z}_i\varepsilon_i) = \mathbf{0}$. The estimator is consistent but generally not unbiased in finite samples.
Two-stage least squares (2SLS)	Generalizes IV to settings with multiple instruments. The first stage projects the endogenous regressors onto the instrument space; the second stage estimates the structural equation using the projected regressors.	$\hat{\boldsymbol{\beta}}_{\text{2SLS}} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y}$, where $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$.	Requires valid and sufficiently strong instruments. In the just-identified case, 2SLS coincides with the simple IV estimator. In the overidentified case, it combines all instruments and is efficient among IV estimators under homoskedasticity.
GMM estimator	Uses the sample analogue of population moment conditions and chooses the parameter vector that makes those sample moments as close to zero as possible, according to a weighting matrix. It nests IV and 2SLS as special cases.	$\hat{\boldsymbol{\beta}}_{\text{GMM}} = \arg\min_{\boldsymbol{\beta}} Q_n(\boldsymbol{\beta})$, $Q_n(\boldsymbol{\beta}) = g_n(\boldsymbol{\beta})'\mathbf{W}_n g_n(\boldsymbol{\beta})$, $g_n(\boldsymbol{\beta}) = n^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$. For the linear model, $\hat{\boldsymbol{\beta}}_{\text{GMM}} = \mathbf{A}^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{y}$, $\mathbf{A} = \mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X}$.	Requires correctly specified moment conditions, identification, convergence of the sample moments, and a positive definite limiting weighting matrix. The efficient GMM estimator uses $\mathbf{W} = \boldsymbol{\Omega}^{-1}$, where $\boldsymbol{\Omega}$ is the asymptotic variance-covariance matrix of the moment conditions.

6. Review Questions and Exercises

Exercise 6.1 (Conceptual foundations)**.** Explain why endogeneity is an identification problem rather than merely a small-sample problem. In your answer, distinguish between sampling variability and asymptotic bias, and explain why increasing the sample size does not generally solve the problem of endogenous regressors.

Exercise 6.2 (OLS under exogeneity)**.** Consider the linear model

$$y_i = \mathbf{x}_i'\boldsymbol{\beta}_0 + \varepsilon_i.$$

Suppose that

$$\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0} \qquad \text{and} \qquad \mathbb{E}[\mathbf{x}_i\mathbf{x}_i']$$

is finite and nonsingular.

(i) Show that

$$\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{\beta}_0 + \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\left(\frac{1}{n}\mathbf{X}'\boldsymbol{\varepsilon}\right).$$

(ii) Use the law of large numbers and Slutsky's theorem to show that

$$\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} \xrightarrow{p} \boldsymbol{\beta}_0.$$

Exercise 6.3 (OLS inconsistency under endogeneity)**.** Suppose the linear model is correctly specified, but

$$\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \boldsymbol{\delta} \neq \mathbf{0}.$$

Show that

$$\operatorname{plim} \hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{\beta}_0 + \{\mathbb{E}(\mathbf{x}_i\mathbf{x}_i')\}^{-1}\boldsymbol{\delta}.$$

Explain why the second term represents asymptotic bias.

Exercise 6.4 (Sources of endogeneity)**.** For each of the following cases, explain why the regressor of interest may be endogenous:

(i) estimating the return to schooling when ability is unobserved;

(ii) estimating demand when price and quantity are jointly determined;

(iii) estimating the effect of income on consumption when income is measured with error;

(iv) estimating the effect of class size on learning outcomes when parents sort into schools.

For each case, suggest one possible instrument and discuss whether it is likely to satisfy relevance and exogeneity.

Exercise 6.5 (Omitted variable bias)**.** Suppose the true model is

$$y_i = \beta_0 + \beta_1 x_i + \gamma c_i + u_i, \qquad \mathbb{E}[u_i \mid x_i, c_i] = 0,$$

but the researcher estimates

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

(i) Write the composite error term ε_i.

(ii) Derive $\mathrm{Cov}(x_i, \varepsilon_i)$.

(iii) Under what conditions is OLS inconsistent?

(iv) Explain how the sign of the omitted variable bias depends on γ and $\mathrm{Cov}(x_i, c_i)$.

Exercise 6.6 (Measurement error)**.** Let the true model be

$$y_i = \beta_0 + \beta_1 x_i^* + u_i,$$

but suppose that the researcher observes

$$x_i = x_i^* + v_i,$$

where v_i is classical measurement error satisfying

$$\mathbb{E}[v_i] = 0, \qquad \mathrm{Cov}(x_i^*, v_i) = 0, \qquad \mathrm{Cov}(u_i, v_i) = 0.$$

(i) Rewrite the model in terms of the observed regressor x_i.

(ii) Show that the composite error is correlated with x_i.

(iii) Derive the probability limit of the OLS estimator of β_1.

(iv) Explain why classical measurement error produces attenuation bias.

Exercise 6.7 (Definition of an instrumental variable)**.** Let z_i be a candidate instrument for an endogenous regressor x_i in the model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

(i) State the relevance condition.

(ii) State the exogeneity condition.

(iii) Explain why relevance alone is not sufficient for consistency.

(iv) Explain why exogeneity alone is not sufficient for identification.

Exercise 6.8 (Simple IV estimator)**.** Consider the just-identified model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

with instrument z_i, where

$$\mathbb{E}[z_i \varepsilon_i] = 0, \qquad \mathrm{Cov}(z_i, x_i) \neq 0.$$

Assume the variables have been demeaned so that no intercept is needed.

(i) Derive the population moment condition used to identify β_1.

(ii) Show that

$$\beta_1 = \frac{\mathbb{E}[z_i y_i]}{\mathbb{E}[z_i x_i]}.$$

(iii) Derive the corresponding sample IV estimator.

(iv) Explain why the estimator is inconsistent if $\mathbb{E}[z_i \varepsilon_i] \neq 0$.

Exercise 6.9 (Matrix IV estimator)**.** Consider the linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

with instrument matrix $\mathbf{Z}$, where the model is just identified so that $\mathbf{Z}'\mathbf{X}$ is square and nonsingular.

(i) Starting from the sample moment condition

$$\frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathrm{IV}}) = \mathbf{0},$$

derive

$$\hat{\boldsymbol{\beta}}_{\mathrm{IV}} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}.$$

(ii) Show that

$$\hat{\boldsymbol{\beta}}_{\mathrm{IV}} - \boldsymbol{\beta}_0 = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\boldsymbol{\varepsilon}.$$

(iii) State the rank and exogeneity conditions needed for consistency.

Exercise 6.10 (First-stage regression and instrument strength)**.** Suppose x_i is endogenous and z_i is proposed as an instrument. The first-stage equation is

$$x_i = \pi_0 + \pi_1 z_i + \mathbf{w}_i'\boldsymbol{\pi}_2 + v_i,$$

where $\mathbf{w}_i$ is a vector of included exogenous controls.

(i) What does the coefficient π_1 measure?

(ii) Why is testing $\pi_1 = 0$ important?

(iii) Explain the meaning of a weak instrument.

(iv) Discuss the consequences of weak instruments for bias, variance, and inference.

Exercise 6.11 (Deriving 2SLS)**.** Let $\mathbf{Z}$ be the matrix of instruments and define the projection matrix

$$\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'.$$

The 2SLS estimator is

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y}.$$

(i) Explain the meaning of the projection matrix $\mathbf{P}_Z$.

(ii) Show how $\mathbf{P}_Z\mathbf{X}$ represents the fitted values from projecting the regressors onto the instrument space.

(iii) Explain why replacing endogenous regressors by their fitted values helps restore consistency.

(iv) Under what condition does 2SLS reduce to the simple IV estimator?

Exercise 6.12 (Consistency of 2SLS)**.** Using

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{y},$$

and

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon},$$

show that

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} - \boldsymbol{\beta}_0 = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\boldsymbol{\varepsilon}.$$

Then express this result in terms of $\mathbf{Z}$ and show the probability limit of the right-hand side under instrument exogeneity and the appropriate rank condition.

Exercise 6.13 (GMM moment conditions)**.** Consider the linear IV/GMM moment condition

$$\mathbb{E}[\mathbf{z}_i(y_i - \mathbf{x}_i'\boldsymbol{\beta}_0)] = \mathbf{0}.$$

(i) Define the sample moment function $g_n(\boldsymbol{\beta})$.

(ii) Define the population moment function $g(\boldsymbol{\beta})$.

(iii) Explain the role of the weighting matrix $\mathbf{W}_n$ in the GMM criterion.

(iv) Explain why different positive definite weighting matrices can produce different GMM estimators.

(v) Explain why all such estimators may be consistent but not equally efficient.

Exercise 6.14 (Closed-form linear GMM estimator)**.** Let

$$g_n(\boldsymbol{\beta}) = \frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

and define

$$Q_n(\boldsymbol{\beta}) = g_n(\boldsymbol{\beta})'\mathbf{W}_n g_n(\boldsymbol{\beta}),$$

where $\mathbf{W}_n$ is symmetric positive definite.

(i) Differentiate $Q_n(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$.

(ii) Set the derivative equal to zero.

(iii) Derive the closed-form linear GMM estimator:

$$\hat{\boldsymbol{\beta}}_{\text{GMM}} = (\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\mathbf{W}_n\mathbf{Z}'\mathbf{y}.$$

(iv) Explain how this estimator relates to 2SLS when

$$\mathbf{W}_n = (\mathbf{Z}'\mathbf{Z})^{-1}.$$

Exercise 6.15 (Optimal GMM weighting matrix)**.** Let

$$\sqrt{n}g_n(\boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}),$$

where $\boldsymbol{\Omega}$ is nonsingular.

(i) State the optimal GMM weighting matrix.

(ii) Show that when $\mathbf{W} = \boldsymbol{\Omega}^{-1}$, the asymptotic variance of the GMM estimator becomes

$$\boldsymbol{\Sigma}_{\text{GMM}} = (\mathbf{G}'\boldsymbol{\Omega}^{-1}\mathbf{G})^{-1}.$$

(iii) Explain in words why moments with larger variance should receive smaller weight.

(iv) Explain why the optimal GMM estimator is efficient only relative to estimators using the same set of moment conditions.

Exercise 6.16 (Diagnostic testing in IV/GMM)**.** For each of the following tests, state the null hypothesis, the alternative hypothesis, and the practical interpretation of rejection:

(i) Durbin–Wu–Hausman test for endogeneity;

(ii) first-stage F-test for instrument relevance;

(iii) Kleibergen–Paap underidentification test;

(iv) Stock–Yogo weak-instrument diagnostic;

(v) Hansen J-test for overidentifying restrictions.

Explain why no single diagnostic test is sufficient to establish that an IV/GMM model is credible.

A. Monte Carlo Simulation: OLS, IV, 2SLS, and GMM under Endogeneity

A.1 Simulation Results

This appendix uses a Monte Carlo experiment to illustrate the finite-sample behaviour of four estimators discussed in the handout: ordinary least squares (OLS), the simple instrumental variables (IV) estimator, two-stage least squares (2SLS), and generalized method of moments (GMM). The purpose is to show how endogeneity affects the distribution of the OLS estimator and how valid instruments can restore consistency. The simulation focuses on the coefficient of the endogenous regressor, β_2, because this is the parameter most directly affected by correlation between the regressor and the structural error term.

The data-generating process is based on the linear model

$$y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i,$$

where x_{2i} is endogenous and x_{3i} is exogenous. The instruments for x_{2i} are z_{1i} and z_{2i}, both of which are constructed to be correlated with x_{2i} but uncorrelated with ε_i. The simulation uses sample size $N = 100$ and $R = 10{,}000$ replications. In each replication, OLS is estimated without instruments, the simple IV estimator uses only z_1, while 2SLS and GMM use both instruments z_1 and z_2. The empirical distribution of $\widehat{\beta}_2$ is then summarized using density plots, means, variances, and biases.

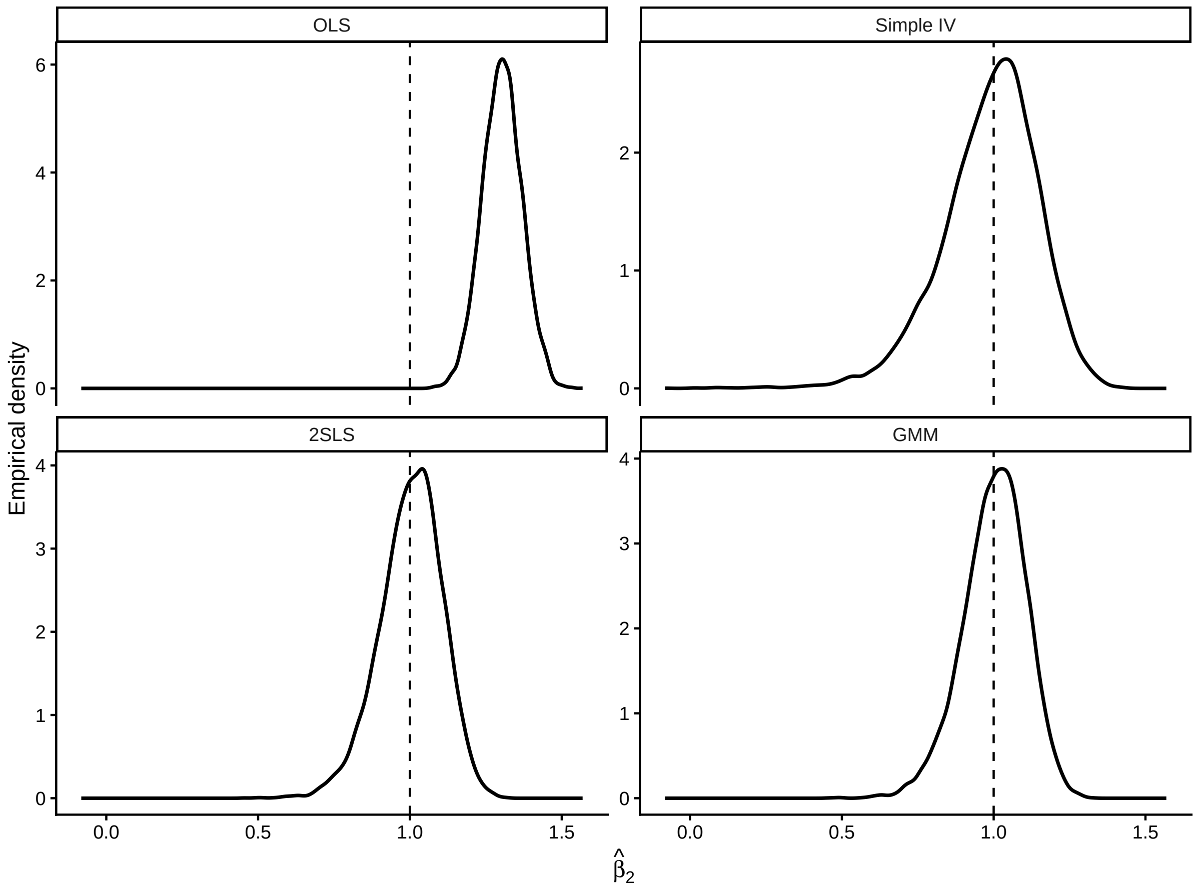

Figure 1: Empirical densities of the Monte Carlo distribution of $\hat{\beta}_2$ for OLS, simple IV, 2SLS, and GMM.

The density plots show the central implication of the simulation. Because x_2 is constructed to be correlated with the structural disturbance ε, the OLS estimator of β_2 is centered away from the true value. This finite-sample pattern is consistent with the analytical probability limit of OLS under endogeneity. By contrast, the simple IV estimator, 2SLS estimator, and GMM estimator are centered much closer to the true value because they use instruments that are correlated with x_2 but uncorrelated with the error term. The simple IV estimator uses only one instrument and is therefore typically more dispersed, while 2SLS and GMM use both instruments and tend to produce tighter empirical distributions. In this design, 2SLS and GMM are expected to behave similarly, although GMM uses a second-step weighting matrix that accounts for the estimated variance-covariance structure of the moment conditions.

Table 2: Mean, variance, and bias of the Monte Carlo distribution of $\hat{\beta}_2$.

Estimator	Mean	Variance	Bias
OLS	1.3033	0.0043	0.3033
Simple IV	0.9869	0.0251	-0.0131
2SLS	1.0025	0.0107	0.0025
GMM	1.0024	0.0108	0.0024

The table summarizes the same lesson numerically. The mean of the OLS estimates is expected to differ substantially from the true value $\beta_2 = 1$, producing a non-negligible bias. This reflects the fact that OLS uses the endogenous variation in x_2, including the component correlated with the disturbance term. The simple IV estimator should display much smaller bias because it uses only exogenous variation from z_1, although its variance may be larger because it discards the additional information contained in z_2. The 2SLS and GMM estimators use both instruments and therefore generally combine low bias with improved precision. The relative performance of 2SLS and GMM depends on the weighting matrix and the variance structure of the moment conditions; in this homoskedastic design, their results are likely to be close, while under heteroskedasticity the efficient GMM estimator may offer clearer efficiency gains.

A.2 R Code

The following code reproduces the Monte Carlo simulation, the density figure, and the summary table reported above.

```
# Load the packages used for simulation, reshaping, plotting, and tables.
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)

set.seed(12345)

# Monte Carlo design
N <- 100
R <- 10000

# True parameters
beta_1 <- 1.0
```

```
beta_2 <- 1.0
beta_3 <- 0.5

# Strength of endogeneity
rho <- 0.60

# Storage
results <- matrix(NA_real_, nrow = R, ncol = 4)
colnames(results) <- c("OLS", "Simple IV", "2SLS", "GMM")

# Define a simple two-step linear GMM estimator.
gmm_linear <- function(y, X, Z) {

  n <- nrow(X)

  # First-step GMM using 2SLS weighting matrix
  W1 <- solve(crossprod(Z) / n)
  beta_1step <- solve(
    crossprod(X, Z) %*% W1 %*% crossprod(Z, X),
    crossprod(X, Z) %*% W1 %*% crossprod(Z, y)
  )

  # Residuals from first step
  uhat <- as.vector(y - X %*% beta_1step)

  # Robust estimate of moment variance
  Zu <- Z * uhat
  S <- crossprod(Zu) / n

  # Small ridge adjustment for numerical stability
  S <- S + diag(1e-8, ncol(S))

  # Second-step optimal weighting matrix
  W2 <- solve(S)

  beta_2step <- solve(
    crossprod(X, Z) %*% W2 %*% crossprod(Z, X),
    crossprod(X, Z) %*% W2 %*% crossprod(Z, y)
```

```
  )

  as.vector(beta_2step)
}

for (r in 1:R) {

  # Instruments and exogenous regressor
  z1 <- rnorm(N)
  z2 <- rnorm(N)
  x3 <- rnorm(N)

  # Components generating endogeneity
  v <- rnorm(N)
  e <- rnorm(N)

  # Structural error correlated with v
  eps <- rho * v + sqrt(1 - rho^2) * e

  # Generate an endogenous regressor using instruments and the latent shock
   ↪  v.
  x2 <- 0.7 * z1 + 0.7 * z2 + 0.5 * x3 + v

  # Outcome variable
  y <- beta_1 + beta_2 * x2 + beta_3 * x3 + eps

  # Matrices
  X <- cbind(1, x2, x3)

  # OLS
  beta_ols <- solve(crossprod(X), crossprod(X, y))

  # Simple IV: just identified, using z1 plus included exogenous regressors.
  Z_iv <- cbind(1, z1, x3)
  beta_iv <- solve(crossprod(Z_iv, X), crossprod(Z_iv, y))

  # 2SLS: use both instruments plus included exogenous regressors.
  Z_all <- cbind(1, z1, z2, x3)
```

```r
  Pz <- Z_all %*% solve(crossprod(Z_all)) %*% t(Z_all)
  beta_2sls <- solve(
    crossprod(X, Pz %*% X),
    crossprod(X, Pz %*% y)
  )

  # Two-step GMM: use the same full instrument set.
  beta_gmm <- gmm_linear(y = y, X = X, Z = Z_all)

  # Store beta_2 estimates
  results[r, ] <- c(beta_ols[2], beta_iv[2], beta_2sls[2], beta_gmm[2])
}

results_df <- as.data.frame(results)
```

```r
# Reshape the simulation output into long form for plotting.
density_df <- results_df |>
  pivot_longer(
    cols = everything(),
    names_to = "Estimator",
    values_to = "beta2_hat"
  )

density_df$Estimator <- factor(
  density_df$Estimator,
  levels = c("OLS", "Simple IV", "2SLS", "GMM")
)

# Plot the empirical distribution of the target coefficient by estimator.
ggplot(density_df, aes(x = beta2_hat)) +
  geom_density(linewidth = 0.8) +
  geom_vline(xintercept = beta_2, linetype = "dashed") +
  facet_wrap(~ Estimator, nrow = 2, ncol = 2, scales = "free_y") +
  labs(
    x = expression(hat(beta)[2]),
    y = "Empirical density"
  ) +
  theme_classic()
```

```r
# Compute Monte Carlo means, variances, and biases by estimator.
summary_table <- density_df |>
  group_by(Estimator) |>
  summarise(
    Mean = mean(beta2_hat),
    Variance = var(beta2_hat),
    Bias = mean(beta2_hat) - beta_2,
    .groups = "drop"
  )

summary_table |>
  mutate(
    Mean = round(Mean, 4),
    Variance = round(Variance, 4),
    Bias = round(Bias, 4)
  ) |>
  kable(
    booktabs = TRUE,
    align = c("l", "r", "r", "r")
  )
```

A.3 Documentation of the R Code

This subsection explains the code presented in Appendix A.2 in the same order in which it is written and executed. The code naturally falls into three parts: the simulation itself, the construction of the density figure, and the construction of the summary table.

The first code block sets up and runs the Monte Carlo experiment. It loads the packages required for plotting, reshaping data, and formatting tables; fixes the random seed for reproducibility; and defines the simulation dimensions $N = 100$ and $R = 10000$. It then specifies the true parameter values β_1, β_2, and β_3, together with the endogeneity parameter ρ. Inside the simulation loop, the code generates two instruments z_1 and z_2, one exogenous regressor x_3, and latent shocks v and e. The structural disturbance ε is constructed from v and e, so that ε is correlated with the endogenous regressor x_2. After generating the outcome variable y, the code estimates the coefficient on x_2 using OLS, a simple just-identified IV estimator, 2SLS, and a two-step GMM estimator. The estimate of β_2 from each method is stored in the results matrix for later comparison.

The second code block turns the simulation output into the figure reported in Appendix A.1. The matrix of simulation results is first converted into a data frame and then reshaped from wide form to long form using `pivot_longer()`. This produces one column for the estimator label and one column for the simulated values of $\hat{\beta}_2$. The estimator labels are then converted into an ordered factor so that the panels appear in a consistent order. Finally, `ggplot()` is used to draw empirical density curves for the four estimators, with a dashed vertical line marking the true value β_2. The `facet_wrap()` call arranges the four densities in a 2×2 layout, making it easy to compare the sampling distributions visually.

The third code block constructs the summary table reported in Appendix A.1. It groups the long-form simulation data by estimator and computes three Monte Carlo summaries for $\hat{\beta}_2$: the mean, the variance, and the bias relative to the true coefficient β_2. The bias is defined as the average estimate minus the true parameter, so it shows whether an estimator is centered on the target. After these quantities are computed, the code rounds them to four decimal places and passes the result to `kable()` with `booktabs` styling, while Quarto supplies the table caption and placement metadata through the chunk options.

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